

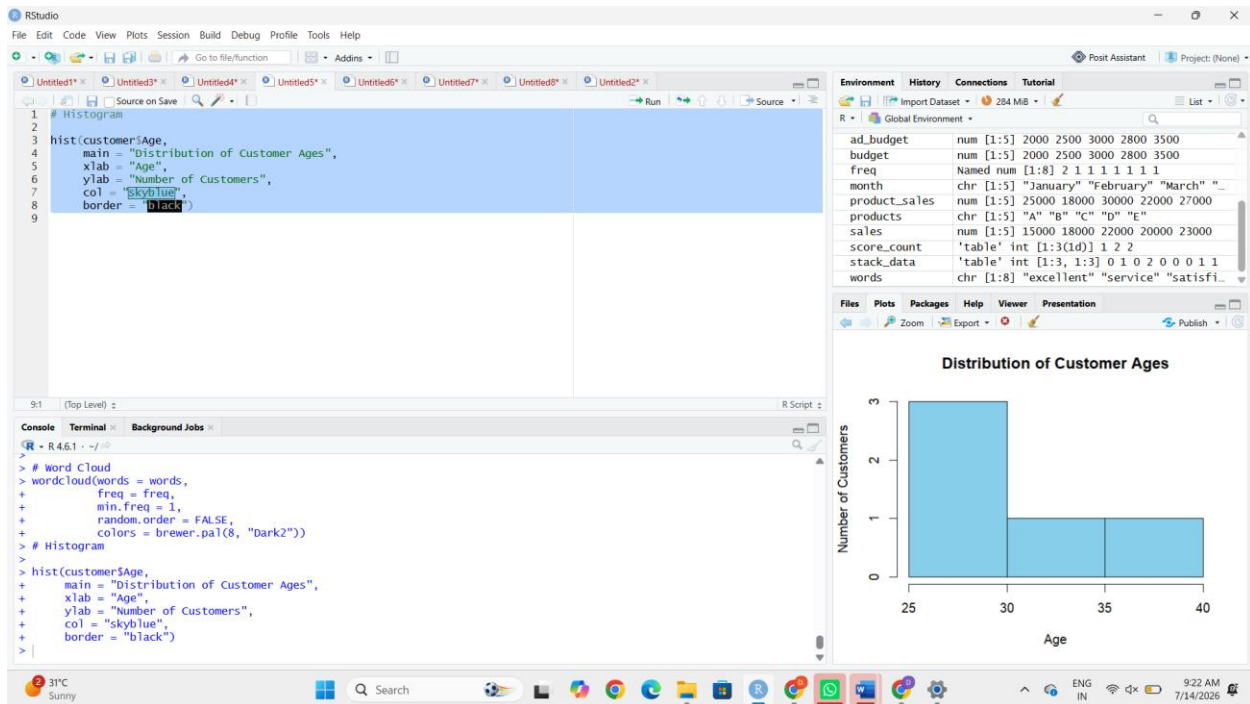
DSA0607-Data Handling and Visualization

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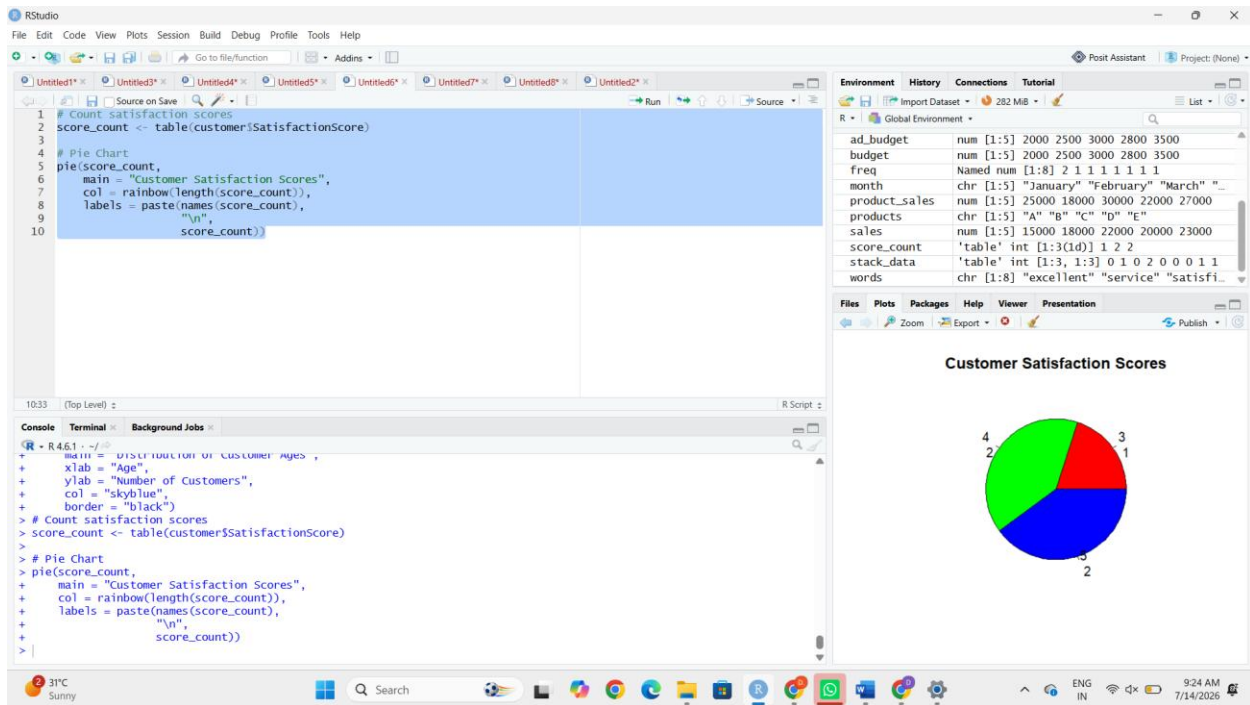
Reg no:192424267

Lab programs

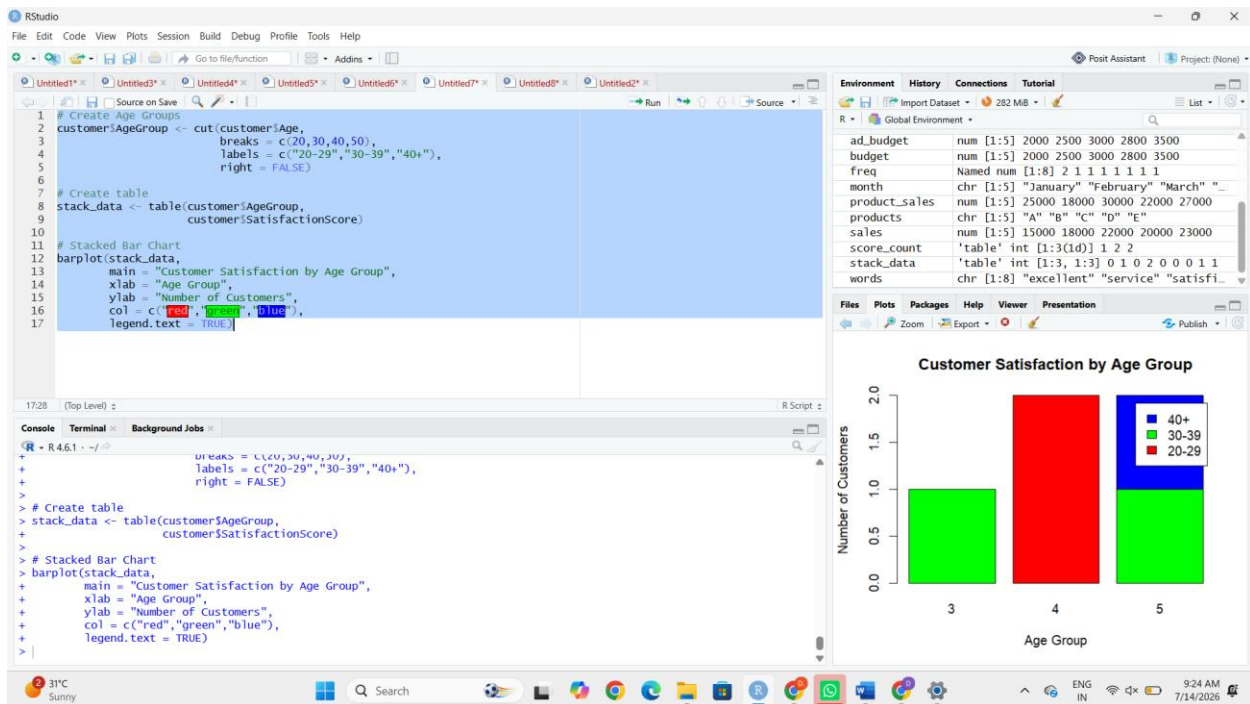
Exp 2



2.2



2.3



2.4

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for text processing and word cloud generation.
- Environment:** Lists variables created in the global environment, including data frames and matrices.
- Console:** Shows the execution output of the R script.
- Word Cloud:** A visualization of the text data, with words sized and colored by frequency.

R Code (Source Editor):

```
9  
10 # Remove numbers  
11 docs <- tm_map(docs, removeNumbers)  
12  
13 # Remove common words  
14 docs <- tm_map(docs, removeWords, stopwords("english"))  
15  
16 # Create term-document matrix  
17 tdm <- TermDocumentMatrix(docs)  
18  
19 m <- as.matrix(tdm)  
20  
21 words <- rownames(m)  
22  
23 freq <- rowSums(m)  
24  
25 # word Cloud  
26 wordcloud(words = words,  
27           freq = freq,  
28           min.freq = 1,  
29           random.order = FALSE,  
30           colors = brewer.pal(8, "Dark2"))
```

Environment:

Variable	Class	Dimensions	Values
ad_budget	num	[1:5]	2000 2500 3000 2800 3500
budget	num	[1:5]	2000 2500 3000 2800 3500
freq	Named num	[1:8]	2 1 1 1 1 1 1 1
month	chr	[1:5]	"January" "February" "March" "..."
product_sales	num	[1:5]	25000 18000 30000 22000 27000
products	chr	[1:5]	"A" "B" "C" "D" "E"
sales	num	[1:5]	15000 18000 22000 20000 23000
score_count	'table' int	[1:3(1d)]	1 2 2
stack_data	'table' int	[1:3, 1:3]	0 1 0 2 0 0 0 1 1
words	chr	[1:8]	"excellent" "service" "satisfi..."

Console:

```
R - R4.6.1 - ~/...  
> # Create term-document matrix  
> tdm <- TermDocumentMatrix(docs)  
>  
> m <- as.matrix(tdm)  
>  
> words <- rownames(m)  
>  
> freq <- rowSums(m)  
>  
> # Word Cloud  
> wordcloud(words = words,  
+           freq = freq,  
+           min.freq = 1,  
+           random.order = FALSE,  
+           colors = brewer.pal(8, "Dark2"))  
>
```

Word Cloud:

The word cloud visualizes the text data. The words are sized and colored by frequency. The words visible are: friendly, service, experience, staff, excellent, satisfied, average, support.